Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/23/2018
PI Name:	Basner, Mathias M.D., Ph.D.		
Project Title:	Cognitive Performance and Crew Cohesion during Confinement in NASA's Human Research Program Human Exploration Research Analog (HERA)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	04/24/2014	**End Date:**	08/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2017 per NSSC information (Ed., 5/16/17) NOTE: Extended to 5/31/2017 per PI (Ed., 2/3/17) NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/17/17) NOTE: Extended to 3/31/2017 per NSSC information (Ed., 10/5/16) NOTE: Extended to 9/30/2016 per information from D. Arias/HRP BHP (Ed., 2/17/16)		
Key Personnel Changes/Previous PI:	Sarah McGuire, PhD has left the research team.		
COI Name (Institution):	Gur, Ruben Ph.D. (University of Pennsylvania, School of Medicine) Dinges, David Ph.D. (University of Pennsylvania, School of Medicine)		
Grant/Contract No.:	NNX14AH98G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	The recently established Human Exploration Research Analog (HERA) plans for campaigns of incremental duration starting early 2014. A campaign is defined as one integrated protocol with one primary mission scenario. A campaign will consist of multiple missions in order to meet study subject requirements. Mission durations are planned from 7 days up to 30 days. The HERA planning schedule currently anticipates 4 missions per year (one per quarter), of 7-day duration in 2014, 14-day duration in 2015, and 28-day duration in 2016. This project seeks to be implemented in 4 missions of campaign 1 (7-day missions) and in 4 missions of campaign 2 (14-day missions). Human exploration of space requires astronauts to maintain a high level of cognitive performance in the presence of a range of fatigue-related conditions associated with perturbations of sleep and circadian neurobiology (e.g., slam shifts), excessive cognitive and physical workloads (e.g., EVAs), and environmental stressors (e.g., elevated CO2, hypoxia, radiation). This highlights the critical need for comprehensive, sensitive, and valid neurocognitive assessments in spaceflight in general and during exploration type missions specifically to guarantee mission success. This project will systematically assess cognitive performance in HERA crewmembers that are confined to the HERA facility for 7-14 days. HERA crewmembers will be exposed to different stressors including high and low workload, sleep deprivation, and time delays. The Cognition test battery consists of 10 validated and brief neuropsychological tests that cover a wide array of cognitive domains and that were specifically developed for high-performing astronauts by the research team. Our research in HERA has the following specific aims: Specific Aim 1: Test feasibility of regular Cognition data acquisition in the space analog environment of HERA. The Cognition software will be installed on Apple iPads. Before pre-isolation baseline testing, the crew will be asked to watch a standardized introductory video that familiarizes them with the Cognition software. They will then perform the battery (i.e., all 10 tests) once for familiarization, and will be asked to perform all practice tests during this first administration. The crew of 4 will then perform Cognition 3 times pre-mission, on a daily basis in-mission (in the 2-hour period pre bed), and 3 times post-mission. The Principal Investigator (PI) and his team will download the data and perform quality control checks shortly after the completion of each Cognition battery. This way, both technical and motivational problems can be addressed immediately. The crew will be de-briefed by the PI and his team by teleconference after the last post-mission test administration. Specific Aim 2: Perform descriptive analysis of changes in cognitive performance with time in mission and related to mission stressors. Key variables of the 10 cognitive performance tests will be analyzed descriptively and plotted by time in mission for each of the 4 HERA missions. Test raw data will be provided to NASA's Life Sciences Data Archive, and thus shared with other investigators. This way, individual investigators can analyze cognitive performance data related to mission events and interventions of specific experiments. Specific Aim 3: Measure rest activity patterns and light exposure with actigraphs (campaign 1 only). The crewmembers will continuously wear wrist actigraphs to determine rest-activity patterns and light intensity throughout the missions. These data will be used to characterize average sleep duration, the placement of sleep periods (naps?), the overall level of activity, and the intensity of light exposure depending on time in mission and mission specific events. Specific Aim 4: Test feasibility of proximity data acquisition in HERA (campaign 1 only). Seven beacons will be strategically placed throughout the facility (3 on floor 1, 3 on floor 2, and 1 in the hygiene module). Each crewmember will wear one beacon and one receiver continuously throughout the mission. We will investigate whether we will be able to detect where in the facility and with whom crewmembers spend most of their time. As this technology is very new, this specific aim is exploratory. Project Methodology. Subjects and Design. For each of the 4 one-week missions, 4 mixed-gender crews will be selected by NASA. The crewmembers will be NASA employees that serve as astronaut surrogates and pass the applicable health screenings. Both NASA and UPenn IRB (Institutional Review Board) approval have been obtained. Cognition test battery: The Cognition test battery consists of 10 brief neuropsychological tests that cover a wide range of cognitive domains (executive control, memory, attention, emotional processing, risk decision making, abstraction, sensorimotor speed) that include and go beyond what is currently measured with NASA's WinSCAT battery. Importantly, brain regions involved in performing each of the Cognition tests have already been established with fMRI, and the tests that are the basis for Cognition have been well-validated in both healthy individuals and patient populations. Cognition can be performed on PCs/laptops running Windows 7 or higher, or on iPads/iPad minis. This assures maximal flexibility in test administration, and no up mass for spaceflight. NOTE: This study concluded in FY2016 ; the grant was augmented several times to allow the PI to provide the Cognition test battery and his expertise in analyzing and interpreting the data to other PIs with related studies.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal. Dr. Basner and his team have developed the Cognition battery specifically for NASA and the astronaut population. Ground and space flight testing of the battery are in the process of data collection. To consider transitioning such a tool to an operational use, further validation and testing is required. This would be a requirement of any new cognitive battery. The highly specialized testing that is offered by the proposed recipient is unique and to date we have not found a cognitive toolkit that provides the level of specificity and usability as the Cognition toolkit. Since all the work has been done my Dr. Basner to date, it is not feasible to begin with a new contractor. The costs to start over would likely match those spent with University of Pennsylvania to date. This is a very low cost effort. The majority of the budget is directed towards labor/personnel which has been deemed reasonable for the amount of hours proposed.
Research Impact/Earth Benefits:	The Cognition test battery was specifically designed for the high performing and highly motivated astronaut population. It will thus be a useful tool for the objective assessment of cognitive performance in similar high performing populations on Earth (e.g., medical doctors).

Task Progress:

NOTE: This study concluded in FY2016 ; the grant was augmented several times to allow the PI to provide the Cognition test battery and his expertise in analyzing and interpreting the data to other PIs with related studies.
This project had the following specific aims:

Specific Aim 1: Test feasibility of regular Cognition data acquisition in the space analog environment of HERA

Specific Aim 2: Perform descriptive analysis of changes in cognitive performance with time in mission and related to mission stressors

Specific Aim 3: Measure rest activity patterns and light exposure with actigraphs (campaign 1 only)

Specific Aim 4: Test feasibility of proximity data acquisition in HERA (campaign 1 only)

Overall, we were able to successfully administer Cognition in HERA, with 97.8% completed tests and with ~1.5% aborted individual tests. This demonstrates feasibility of deploying Cognition in the space analog environments like HERA. At the same time, many valuable lessons were learned relative to Internet connectivity, feedback scores, and instructions for the subjects relative to using the touch interface.

The only statistically significant effect of age, sex, and educational attainment on key Cognition speed and accuracy test outcomes was seen for the Visual Object Learning (VOLT), where the time needed for subjects to complete the test increased with age. It is likely that we were underpowered to detect some of the effects that were described for the Cognition tests in the literature1 or that were observed in the National Space Biomedical Research Institute's (NSBRI) basic Cognition validation study (NSBRI NBPF00009). [see that project for reporting https://]

With increasing time in mission, speed on all Cognition tests increased while accuracy remained unchanged. This practice effect can therefore best be described as “getting faster without sacrificing accuracy.” There was only one exception to this pattern: Speed on the F2B did not increase significantly with time in mission. As HERA is an operational environment, it is unclear to what extent these practice effects can be explained by the specifics of the mission simulation. For example, sleep time increased slightly with time in mission during campaign 1, and feelings of tiredness and sleepiness decreased at the same time. In a normative study performed in astronauts, astronaut candidates, and mission controllers at Johnson Space Center (JSC) we did see increases in accuracy across test administrations (1). However, the largest effects were seen during the first 3 administrations which were excluded from this analysis.

Subjective ratings of tiredness, sleepiness, physical exhaustion, mental fatigue, stress, and workload significantly decreased, while ratings of sleep quality and happiness significantly increased with time in mission. In general, crewmembers rated themselves happy, healthy, not stressed, not depressed, not bored, not lonely, with low levels of monotony and high levels of sleep quality. Ratings of tiredness, workload, sleepiness, physical exhaustion, and mental fatigue were medium to high, but decreased with time in mission. Overall, the observed pattern suggests that, on average, crewmembers coped well with or adjusted well to the isolated and confined HERA environment with increasing time in mission.

Subjects were sleep deprived for part of one night (campaign 1) or a full night (campaign 2). We found significant sleep deprivation effects on Psychomotor Vigilance (PVT) response speed and lapses, Digit Symbol Substitution (DSST) speed, and the ability to correctly identify emotions on the Emotion Recognition (ERT). We also found significant differences between sleep deprivation conditions for the MP, F2B, and AM, but these effects were less consistent. The PVT and DSST were already identified as the two tests with the highest sensitivity to sleep loss, both in general (2) as well as on the Cognition test battery specifically (1). The ERT was the only test to show significant performance decrements in both HERA campaigns. We did not see this effect on the ERT in a sample of 44 subjects undergoing one night of total sleep deprivation (1). However, in this study, subjects performed Cognition shortly after 11 AM when the circadian system has started to promote alertness. The fact that sleep deprivation can affect emotion recognition has been reported in the literature (3,4). We also found significant sleep deprivation effects for subjective ratings of tiredness, sleepiness, physical exhaustion, mental fatigue, sleep quality, happiness and workload, whereas ratings of health, stress, depression, boredom, loneliness, and monotony were unaffected.

Structured debriefs with the crew provided important information on ways how the Cognition software could be further improved (e.g., through the optimization of feedback scores). We also identified most and least favorite tests, and gained important insights on reasons for the likes and dislikes that can be used to increase user acceptability. In general, however, subjects praised the software for its appeal, ease of use, and reliability. Most crewmembers thought that an administration interval of once per week would be optimal for 6-month International Space Station (ISS) missions.

Crewmembers wore two wrist actigraphs 24/7 during the isolation period in campaign 1 only. These measured movement activity, rest-activity patterns and light intensity. Collecting actigraphy data in HERA was feasible with 96.4% complete data during the day and the night. Sleep parameters did not differ between the two watches worn by the same crewmember.

Aside from the sleep deprivation effect on mission day 3, total sleep time, wake after sleep onset, and sleep efficiency differed only marginally between mission days. Objectively assessed total sleep time and subjectively assessed sleep time agreed well on average, but differed up to ±1.5 hours on individual nights. Activity levels decreased from mission day 2 through mission day 6, and then increased significantly on mission day 7. While sampling activity levels was feasible in the HERA environment, the light sensors of the actigraphs were either not sensitive enough or covered too often by long sleeves to provide useful information.

The actiwatches also provided proximity information for each crewmember relative to the other crewmembers and relative to beacons placed strategically throughout the facility. Overall, it was feasible to collect proximity data in the HERA environment. Proximity data replicated that mission specialists 1 and 2, on the one hand, and the flight engineer and commander, on the other hand, were scheduled to spend more time together. Also, the devices provided useful insights into the use of the HERA facility across the 24 hour day.

Low-energy Bluetooth receivers and beacons could be used to both detect when crewmembers are in proximity of one another, and to determine time spent in different locations throughout the facility. Further validation of this approach is needed by comparing measurements based on proximity to video of crew interactions and movement. In the meantime, the firmware of the proximity devices was upgraded by the company, and each device now functions both as a beacon and a receiver. Therefore, each crewmember only needs to wear a single device. This technology is currently used in the Antarctic research stations Concordia and Neumayer by this research team.

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